Central Tendency II

Special averages (geometric and harmonic means) and positional averages (quartiles, deciles, percentiles).

Authors: **Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul**

STAT 3202 · #KAUMeLON · Official revision sheet

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1 Geometric mean (GM)

The nth root of the product of n positive observations. Use it for **growth rates, ratios, percentages** and multiplicative change.

Definition

$$GM = \sqrt[n]{\prod x_i}$$

Using logs

$$GM = \text{Antilog}\left(\frac{\sum \log x_i}{n}\right)$$

Frequency distribution

$$GM = \text{Antilog}\left(\frac{\sum f_i \log x_i}{n}\right)$$

Needs positive values: a zero makes the product GM zero, and it is unsuitable for negatives.

2 Harmonic mean (HM)

The **reciprocal of the arithmetic mean of the reciprocals**. Use it for **rates**, such as average speed over equal distances.

Raw data

$$HM = \frac{n}{\sum \frac{1}{x_i}}$$

Frequency distribution

$$HM = \frac{n}{\sum \frac{f_i}{x_i}}$$

GM and HM both give **more weight to smaller values**.

3 AM · GM · HM

Together: the **Pythagorean means**.

Positive observations

$$AM \geq GM \geq HM$$

Equal only when all observations are equal.

Two positive values

$$AM \times HM = GM^2$$

4 Which mean?

Mean	Use for	Agri example
AM	Additive quantities, general averages	Average yield, rainfall
GM	Growth factors, ratios, percentages	Compound growth of area or production
HM	Rates and reciprocal relationships	Average speed of a tractor over equal distances

5 Positional averages

Measure	Divides into	Key equalities
Quartiles Q_1, Q_2, Q_3	4 parts	Q_2 = Median
Deciles $D_1 \dots D_9$	10 parts	D_5 = Median
Percentiles $P_1 \dots P_{99}$	100 parts	$P_{25} = Q_1,\ P_{50} = Q_2,\ P_{75} = Q_3$

Raw data: **sort first**. Discrete data: use cumulative frequencies. Grouped data: find the class, then interpolate.

6 Raw data positions

First quartile

$$Q_1 = \left(\tfrac{n+1}{4}\right)^{\text{th}}$$

Third quartile

$$Q_3 = \left(\tfrac{3(n+1)}{4}\right)^{\text{th}}$$

dth decile

$$D_d = \left(\tfrac{d(n+1)}{10}\right)^{\text{th}}$$

pth percentile

$$P_p = \left(\tfrac{p(n+1)}{100}\right)^{\text{th}}$$

7 Grouped data (interpolation)

Quartile Q_1

$$Q_1 = l_1 + \frac{\frac{n}{4} - m_1}{f_1}\, c_1$$

Quartile Q_3

$$Q_3 = l_3 + \frac{\frac{3n}{4} - m_3}{f_3}\, c_3$$

Decile

$$D_d = l + \frac{\frac{dn}{10} - cf}{f}\, c$$

Percentile

$$P_p = l + \frac{\frac{pn}{100} - cf}{f}\, c$$

l lower boundary of the class holding the position · m or cf cumulative frequency before it · f its frequency · c its width.

! Common mistakes

- × Forgetting the **antilog** at the end of a GM.
- × Using GM or HM with zero or negative values.
- × AM for average speed over equal distances (use HM).
- × Finding quartiles before sorting.

? Fill in the blanks

1. A tractor ploughs two equal distances at 4 km/h and 6 km/h; its average speed is _____ km/h. **[4.8 (harmonic mean)]**
2. Area grew by factors 2 and 8 in two successive years; the average growth factor is _____. **[4 (geometric mean)]**
3. Two positive numbers have AM = 10 and HM = 6.4, so their GM is _____. **[8]**
4. The 50th percentile, the 5th decile and the 2nd quartile are all equal to the _____. **[median]**
5. The 75th percentile is the same as the _____ quartile. **[third]**
6. The geometric mean cannot be used if any observation is _____. **[zero or negative]**

Source: *Textbook of Agricultural Statistics*, Ch. 5 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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